

JOHNSON REBEIRO

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Backend Software Engineer with 3+ years of experience building scalable APIs, real-time systems, and data-driven platforms. Specialized in Python, distributed systems, and high-performance backend architecture, using Python, Django, PostgreSQL, Redis, Kafka, Docker, and AWS.

Professional Experience

Software Developer Engineer (SDE)

March 2023 - Present

Learn Basics (Lighting Bees Private Limited)

Software Developer Intern Aug 2022 – Mar 2023

- Architected a distributed examination-processing platform spanning 5+ servers and multiple cloud providers, using Redis-backed queues and workers to process 10,000+ images daily through document conversion, bar-code detection, orientation correction, AI question-region detection, answer cropping, and response processing; its Class 10 workflows are trusted by 15+ schools.
- Designed and developed a centralized smart print and scan management platform that intelligently routed print jobs across 8 printers using custom scheduling algorithms and processed real-time barcode events from 4 serial-connected scanners to automate document tracking.
- Built a Neo4j-based knowledge graph integrated with LLMs to power personalized content recommendations across the LMS.
- Led the development of a real-time content management system with multi-user collaboration.
- Built an adaptive worksheet recommendation engine that analysed historical student performance to generate personalized question sets based on learning gaps and proficiency.
- Designed and implemented Docker-based CI/CD pipelines, PostgreSQL replication, server monitoring, and automated alerting to improve deployment reliability and system availability.
- Designed and implemented an automated LaTeX document generation pipeline, replacing manual Power-Point/Word workflows and improving PDF generation performance by 10× while enabling scalable distribution through Amazon S3.
- Developed event-driven backend integrations to synchronize Overleaf, SharePoint, Excel, and LMS data into a centralized PostgreSQL database.
- Enhanced database schema design to support evolving product requirements.
- Automated email and discord messages, reducing manual communication efforts.

Projects

1. Distributed Algorithmic Trading Platform

- Designed and developed a distributed trading platform with real-time market data, order management, and portfolio tracking using Python, FastAPI, PostgreSQL, Redis, Kafka, and Docker.
- Implemented low-latency order processing and Web Socket-based real-time updates using an event-driven architecture.
- Built an order life cycle workflow supporting validation, execution, cancellation, position updates, and persistent trade history.
- Used Redis for caching and Kafka for asynchronous market and order events, enabling scalable processing across independent services.
- Added idempotent event handling, retry mechanisms, and structured logging to improve reliability and prevent duplicate order processing.

Skills

- **Languages:** Python, LaTeX.
- **Frameworks:** Django, FastAPI
- **Databases:** PostgreSQL, Redis, Neo4j, MySQL
- **Cloud & DevOps:** Docker, AWS (EC2, S3), CI/CD, MinIO
- **Messaging:** Kafka, WebSockets
- **Tools:** Git, GitHub, GraphQL, PgAdmin4

Education

Bachelor of Engineering (Electronics & Communication Engineering) 2019 – 2023 | CGPA: 7.61/10
Yenepoya Institute of Technology, Mangalore, India